LESSON PLAN

BASIC RADIO & AVIONICS (25)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	7	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Capacitor type fuel quantity indicator, angle of attack indicator, gyroscopic instruments attitude indicator, turn and slip indicator, direction indicator
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Capacitor type fuel quantity indicator, angle of attack indicator, gyroscopic instruments attitude indicator, turn and slip indicator, direction indicator. Practical on various instruments.
5. Specific Objectives	25.1.1.1 Trainee will be able to understand basic instrument gauges, instrument type auto pilot and maintenance.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Practical on various instruments: Fuel quantity gauge, Angle of Attack indicator, Attitude indicator, Turn and slip indicator, Direction indicator. Safety: Ensure instruments are powered off before handling. Instructor supervision: Monitor trainee interaction with instrument panels.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a gyroscopic instrument?
(b) How does a capacitor type fuel quantity indicator measure fuel levels?
(c) Why is the direction indicator essential for navigation?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Capacitor type fuel quantity indicator operation • Angle of attack indicator principles • Gyroscopic instruments overview	L & Ds	PPT, IB	15 min
(b)	• Attitude indicator construction and function • Turn and slip indicator mechanics • Direction indicator operation	L & Ds	PPT, PDF	20 min
(c)	• Instrument maintenance basics • Troubleshooting common instrument failures • Auto pilot instrument integration	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between the attitude indicator and the turn and slip indicator.
(b) Describe the role of the angle of attack indicator in flight safety.
(c) What are the key maintenance checks for gyroscopic instruments?

12. Home Assignment and Follow Up	Prepare a brief report on the need for a starter and ignition system, linking it to the engine starting system covered in LP 04.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction, need for starter, ignition, brief description of a typical engine starting system. Practical on Engine Starting system.	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min